

How can we measure Nadal’s dominance at Wimbledon?

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Abstract

This study explores a possible method that could be used to determine the greatest tennis player of all time through dominance at a particular tournament. We want to be able to determine Nadal’s strength at Wimbledon over time. I used ggplot in R with publicly available tennis data to create two scatterplot models that measure Nadal’s performance at Wimbledon throughout his current career. The data shows that in recent years, Nadal has become more dominant at Wimbledon, where **dominance is defined to be the “ease” in which one wins a match**. From the data, we can conclude that Nadal’s rise in dominance stems from the improvements in his serve.

Introduction

As of April 2022, Nadal has 21 grand slams (2 Wimbledon) and Djokovic (6 Wimbledon) and Federer (8 Wimbledon) have 20 each. All players’ accomplishments are at the very highest in tennis and too close together to determine the clear “greatest tennis player of all time”. By looking at the level of dominance Nadal held over Wimbledon in his matches compared to Djokovic and Federer, we can have a clearer assessment of Nadal’s level at Wimbledon, which is vital to determine the overarching question of who the greatest tennis player of all time is.

Nadal has historically had struggles on grass, the surface at Wimbledon, due to grass's property of nullifying the spin of the ball, which is Nadal’s greatest strength. The playstyles of Djokovic and Federer, on the other hand, are more suited for grass, and they have won the tournament more times, as a result. Nadal’s playstyle therefore factors into his level of dominance in all of his matches at Wimbledon, which is what this study investigates.

Methods

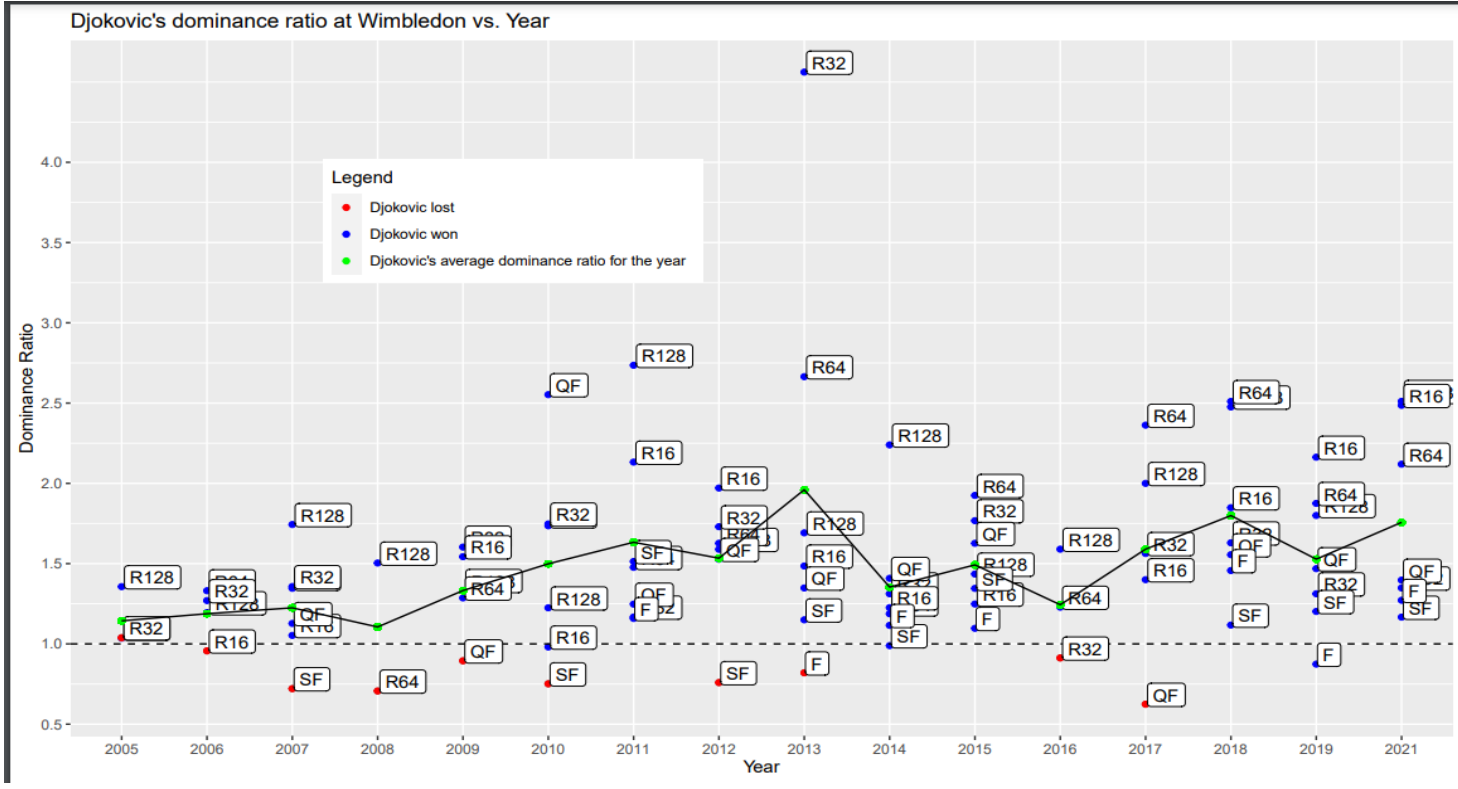
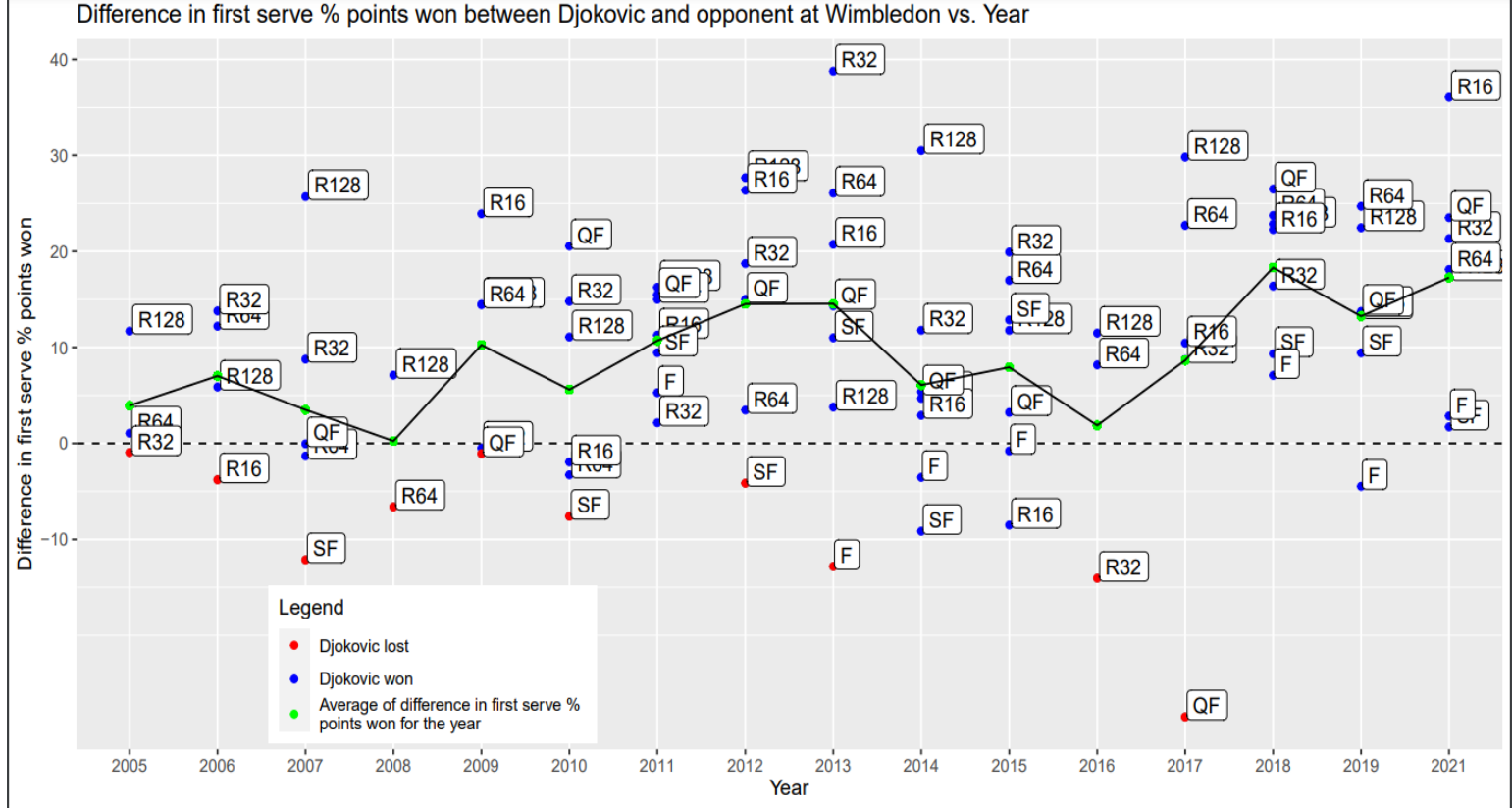
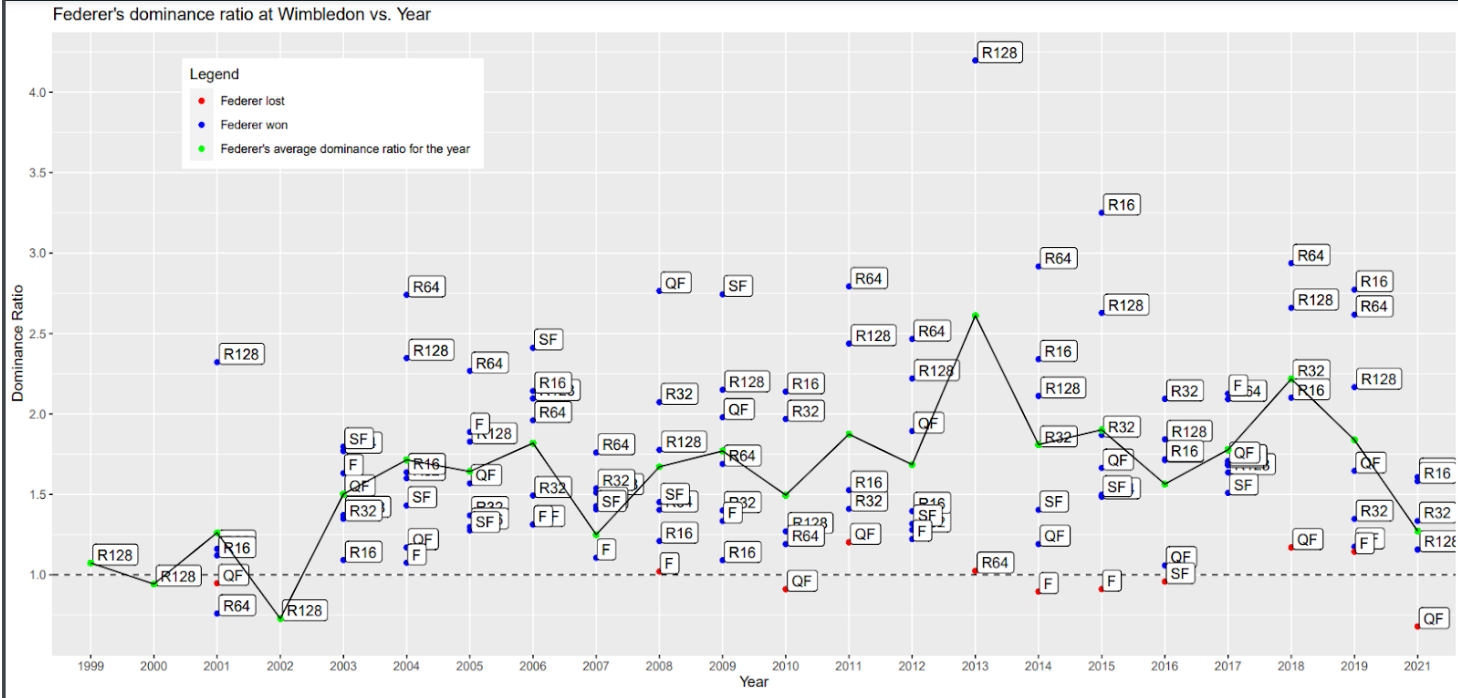
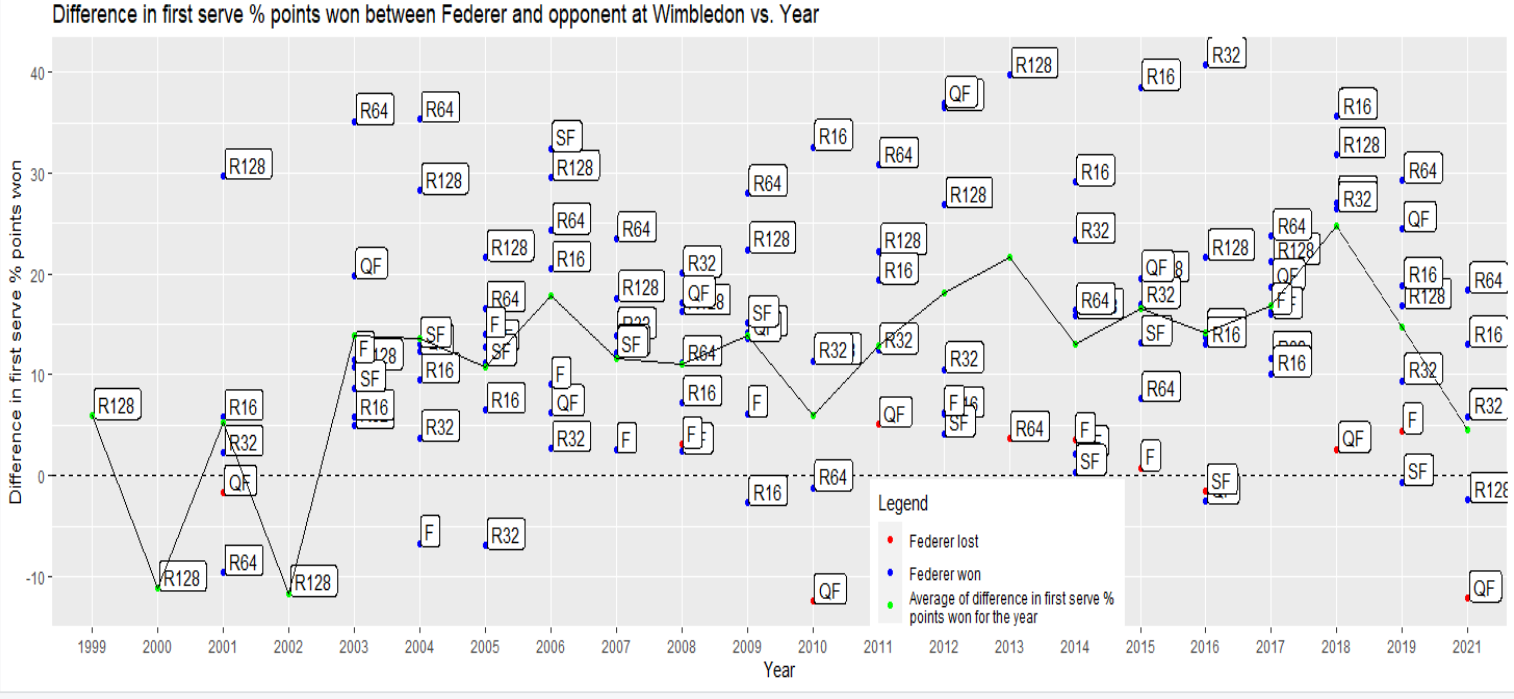
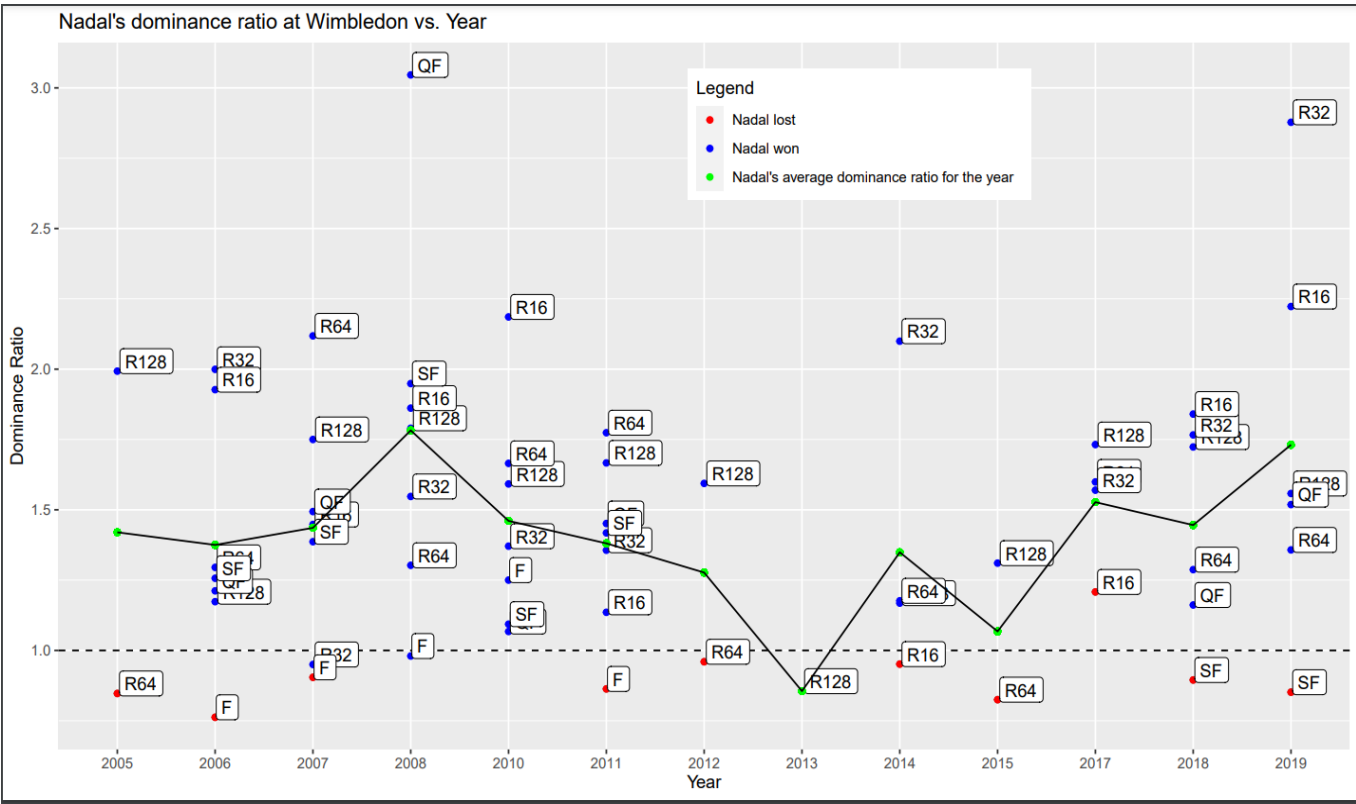
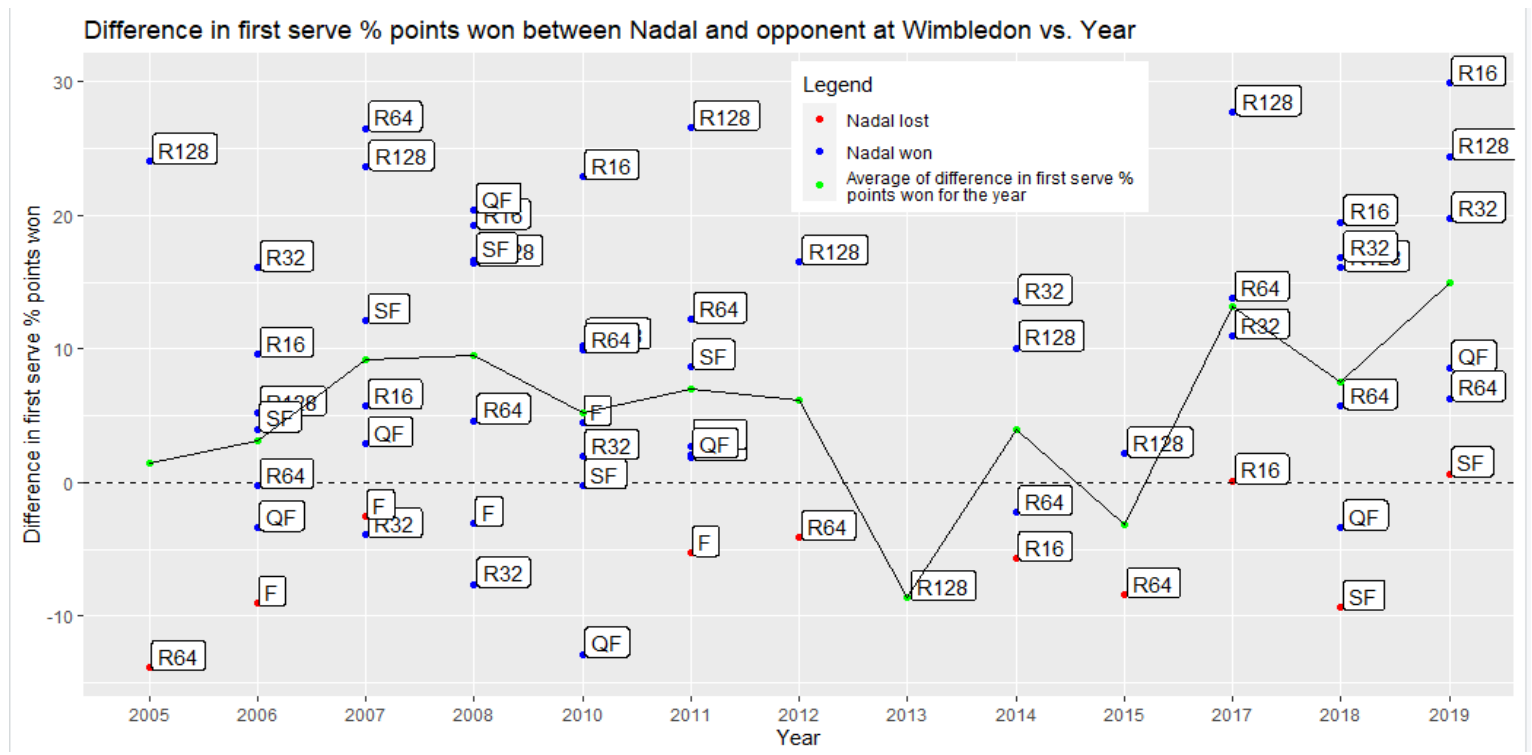
In this study, I chose two models to measure Nadal’s dominance at Wimbledon, difference in % of first serve points won and dominance ratio. A serve (typically overhead) is a shot that is used to start the point and a return is the shot that is used to return the serve.

Difference in % of first serve points won is defined to be (% of the player's first serve points won – % of the opponent's first serve points won). This gives us a relative measure of how dominant a player is on their serve compared to their opponent. If the difference is positive, then the player was more dominant on first serves in the match and if the difference is negative, then the opponent was more dominant on first serves overall in the match.

Dominance ratio is defined as (% of points won on opponent’s serve / % of points lost on own serve). It includes return statistics and tells us how dominant a player is over the course of the entire match. From the dominance ratio, we can make conclusions of a player’s overall performance over a match rather than only on their first serve.

I extracted data from Jeff Sackmann’s publicly available tennis dataset using R, and I used ggplot to create the scatter plot graphs.

Charts



Results and Discussion

We can classify 3 periods in Nadal’s history at Wimbledon: 2006-2011, 2012-2015, and 2017- 2019 (present) 2006-2011 (excluding 2009 due to injury) are his “earlier” years, where he found great success at Wimbledon, making the finals or winning the tournament in these given years. Only players he lost to in this time period were Federer and Djokovic, both of whom are considered to be the best grass players of this decade. Nadal’s matches in 2012 to 2015 are characterized by numerous, seemingly “random” losses to lower-ranked players. During these matches, Nadal was also not physically in form for most of these matches. 2017 to 2019 (present) are the "resurgent" years where Nadal appears to have found good form on grass again, making deep runs.

In spite of consistently making deep runs from 2006 - 2011, there were still a lot of wins he had in which the first serve percentage difference in points won was in the opponent’s favor, an indication Nadal was constantly facing pressure from the opponent on his serve. Federer and Djokovic, on the other hand, rarely played winning matches at Wimbledon where they had a smaller first serve % points won than their opponent, an indication that Djokovic and Federer are more “comfortable” in their service games than Nadal.

From the years 2017-2019, the difference between first serve % points won is higher than it ever has been for Nadal, which therefore indicates a significant improvement in Nadal’s serving performance. Specifically in 2019, Nadal had become so dominant on serve that he even lost a match despite having a higher first serve % points won than his opponent, who was Federer. Due to the increase in first serve % points won present in the last 3 years, we can conclude that Nadal has improved his first serve, which is a major factor that led to his return to making deep runs at Wimbledon.

If we are to compare Nadal’s dominance ratio to that of Federer and Djokovic throughout their careers, we see that Nadal’s average dominance ratio of 1.39 is lower throughout the years than Djokovic’s dominance ratio of 1.46 and Federer’s dominance ratio of 1.61. The dominance ratio tells us a similar story in that Nadal has more difficulty in winning a match at Wimbledon compared to Djokovic and Federer. The main difference is that dominance ratio accounts for % of second serve points won and % of serve return points won. In the years from 2017-2019, Nadal’s average dominance ratio was 1.57, an indication of Nadal’s improvement on grass.

If we use the data from both the difference in first serve % of points won and in dominance ratio, we can make assumptions for the server-returned dynamic in certain matches. For example in the 2019 Wimbledon semifinals between Nadal and Federer, Nadal had a higher % of first serve points won, but Federer had a higher dominance ratio. Therefore, we can assume that Federer was returning serve well, relative to Nadal.

Works Cited:

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